



UNDERSTANDING JAVA FUNDAMENTALS IN 5 MINUTES

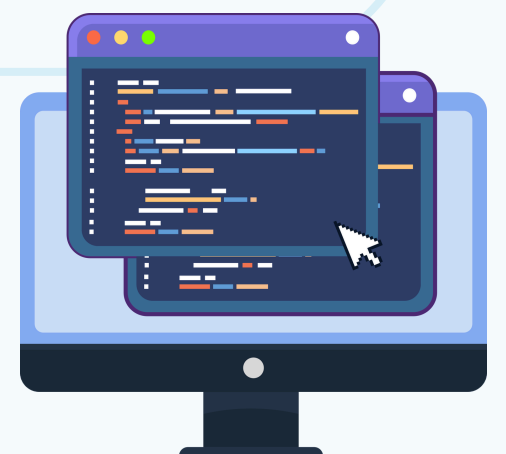
JAVA BASICS:



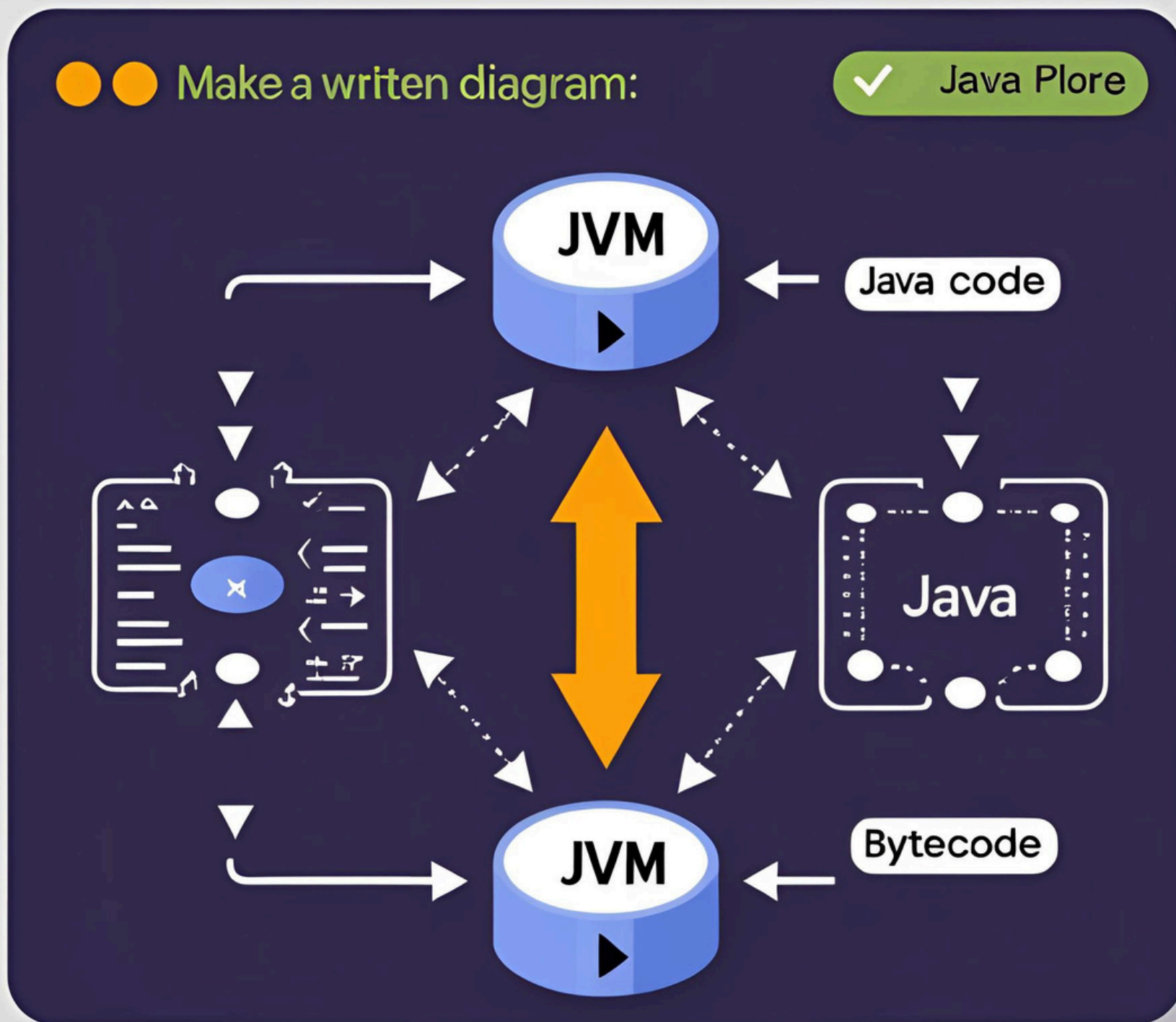
- Object-Oriented Programming Language
- Platform Independent (JVM)
- Strongly Type and Secure

CORE COMPONENTS: </>

- JVM: Executes Java bytecode
- JDK: Development tools
- JRE: Runtime environment



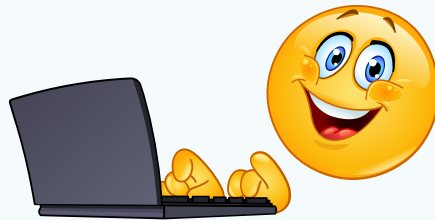
JAVA COMPILATION FLOW



Makes bytecode on any platform



JAVA VIRTUAL MACHINE (JVM):

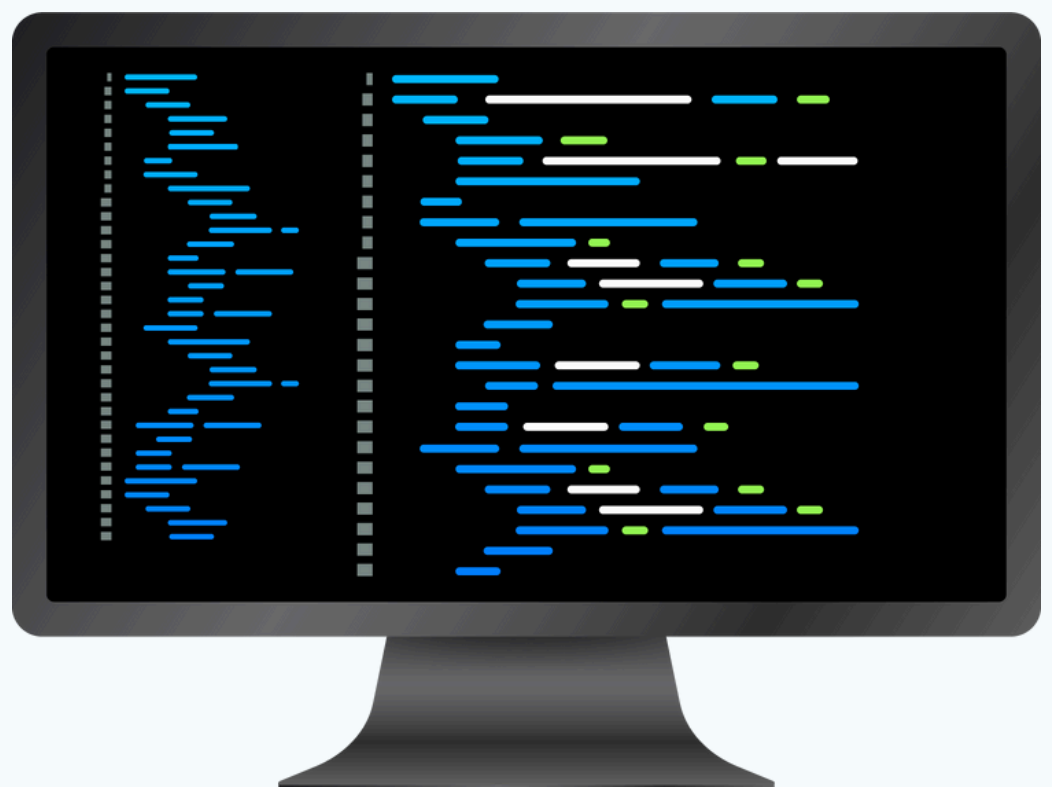


- Loads, verifies and executes bytecode
- Provides garbage collection
- Ensures platform independence

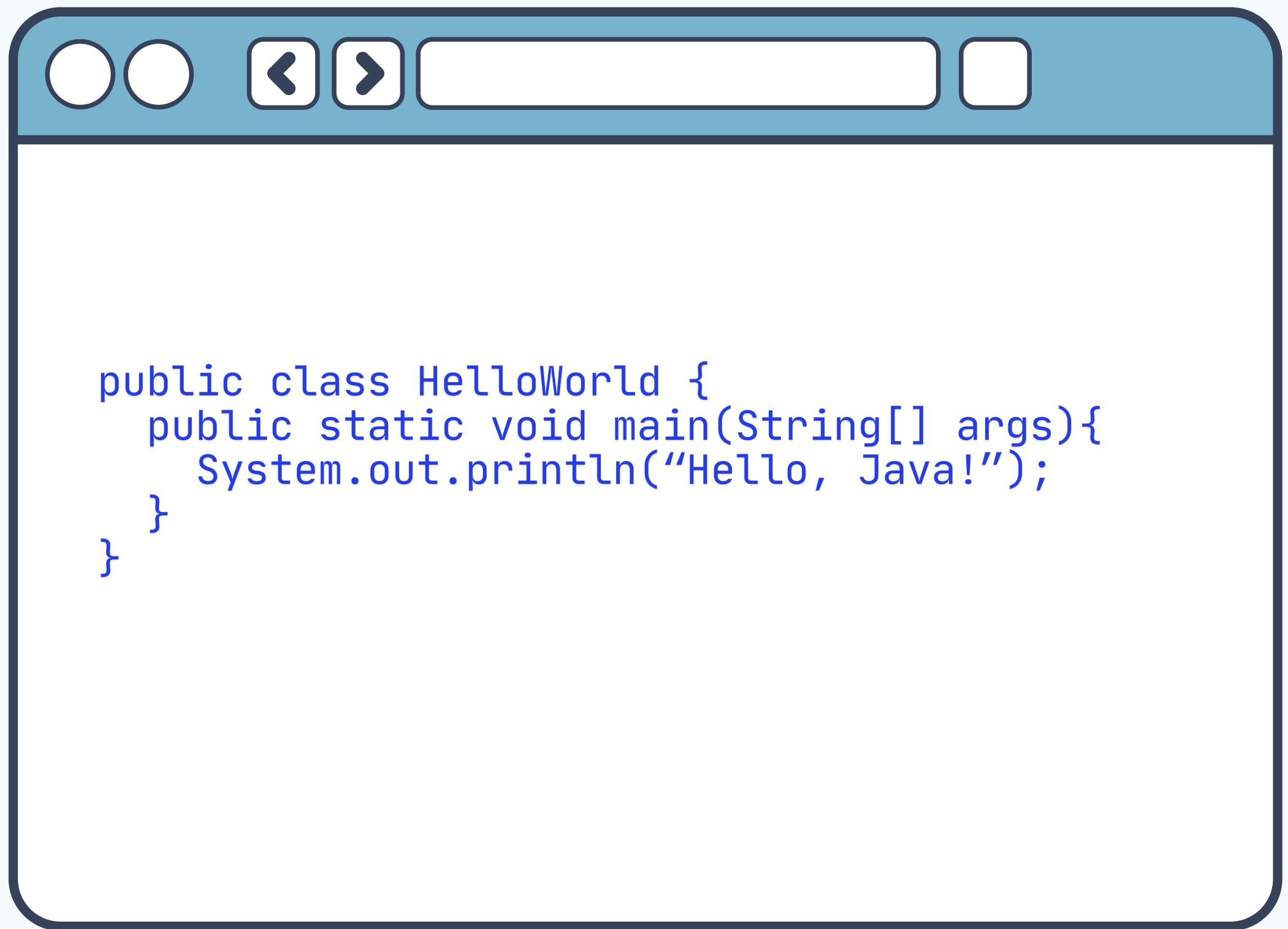
GARBAGE COLLECTION:



- Automatic memory management
- Removes unused objects
- Prevents memory leaks



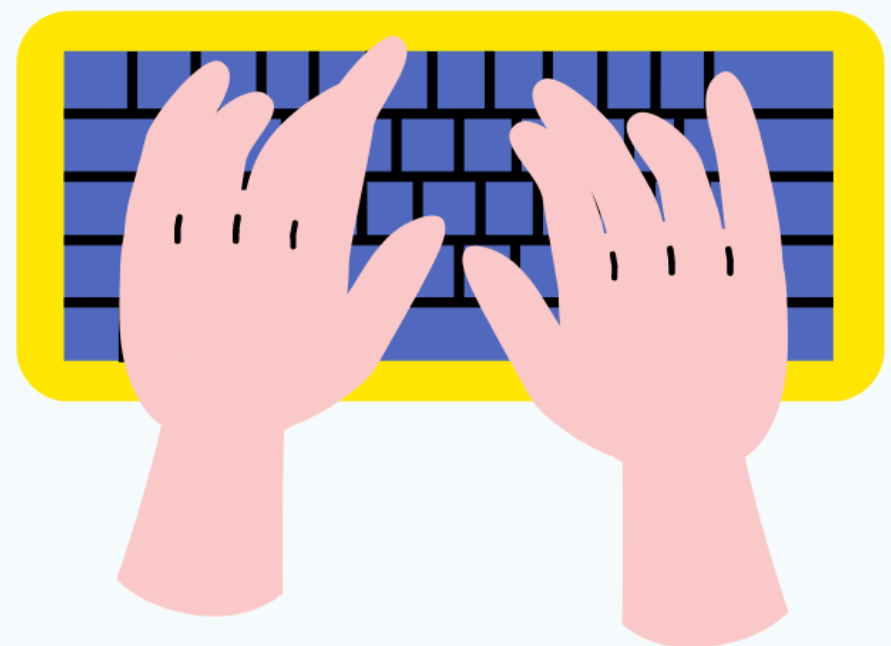
JAVA SYNTAX EXAMPLE



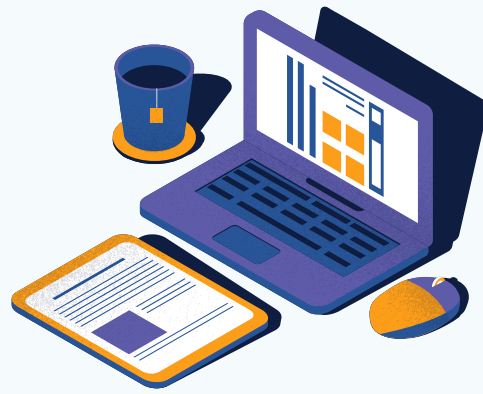
DATA TYPES:



- **Primitive:** int, double, char, boolean
- **Non-Primitive:** String, Arrays, Objects

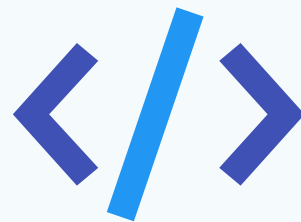


CONTROL FLOW:

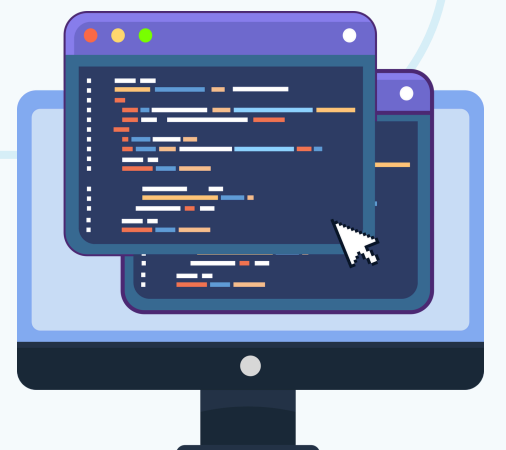


- if / else, switch
- Loops: for, while, do-while

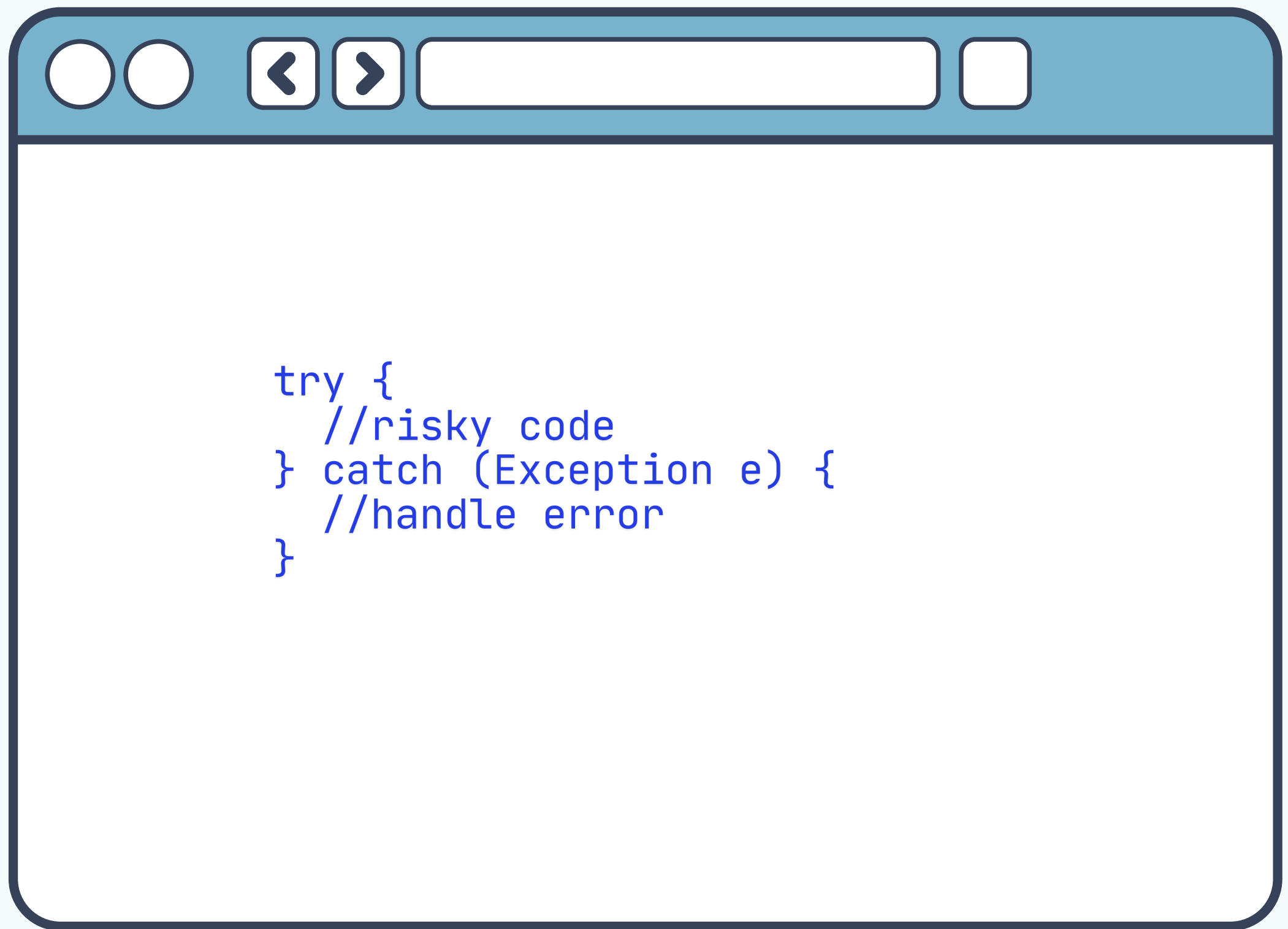
OOP PRINCIPLES:



- Encapsulation
- Inheritance
- Polymorphism
- Abstraction



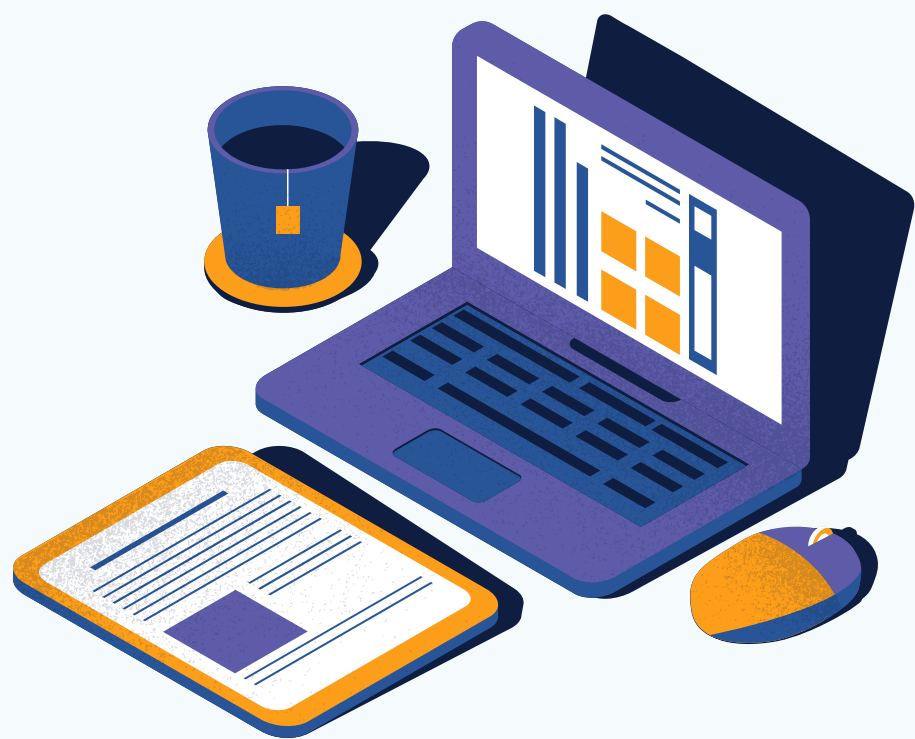
EXCEPTION HANDLING



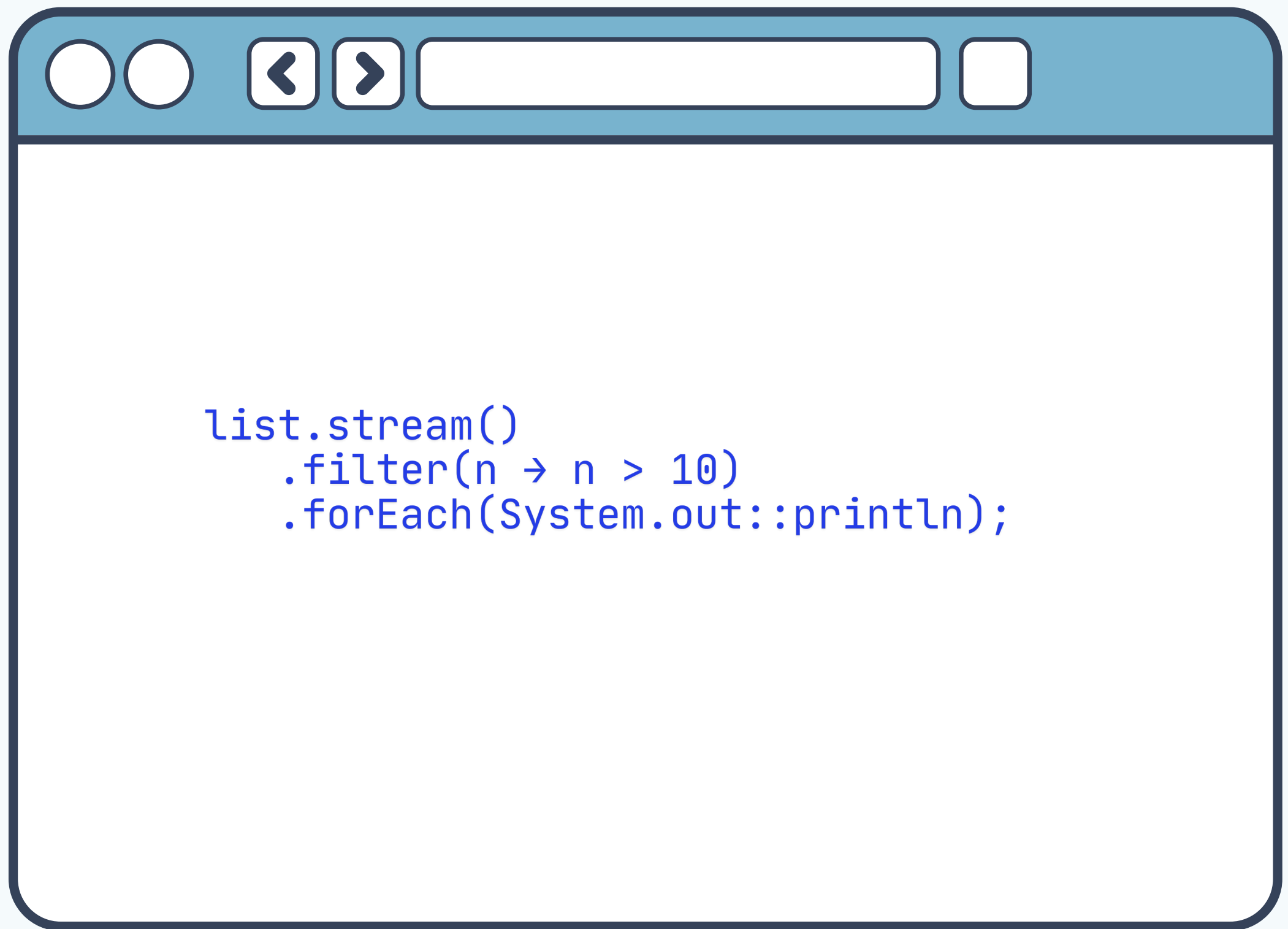
COLLECTIONS FRAMEWORK:



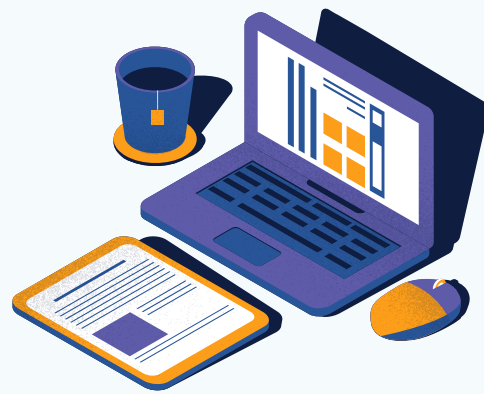
- Interfaces: List, Set, Map
- Implementations: ArrayList, HashSet, HashMap



STREAM API

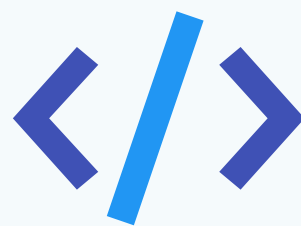


JAVA IN THE REAL WORLD:

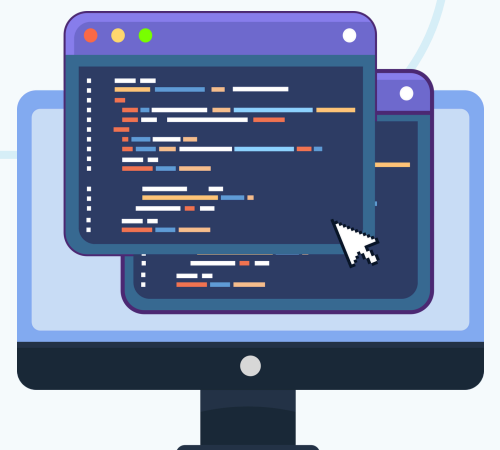


- **Web: Spring Boot**
- **Mobile: Android**
- **Enterprise: Banking systems**

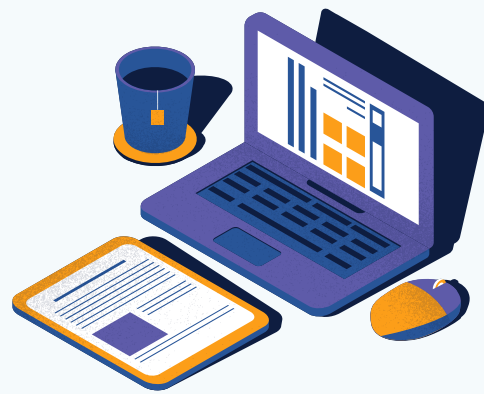
BEST PRACTICES:



- **Write clean, readable code**
- **Use meaningful names**
- **Handle exceptions properly**
- **Keep methods small**



COMMON PITFALLS:

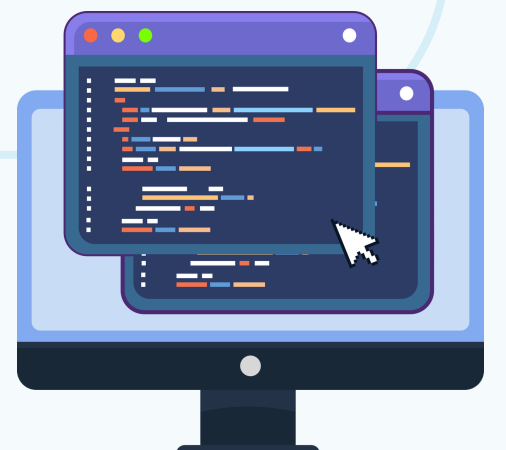


- **NullPointerException**
- **Not closing resources**
- **Overcomplicating logic**

WHERE TO LEARN MORE:



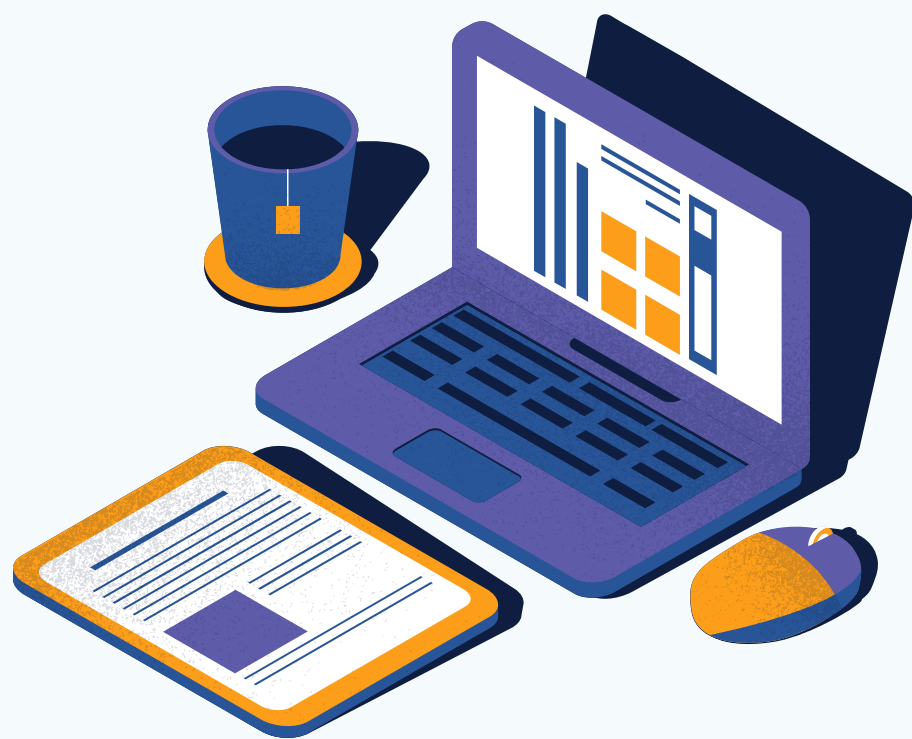
- **Oracle Documentation**
- **Baeldung**
- **Effective Java (Book)**



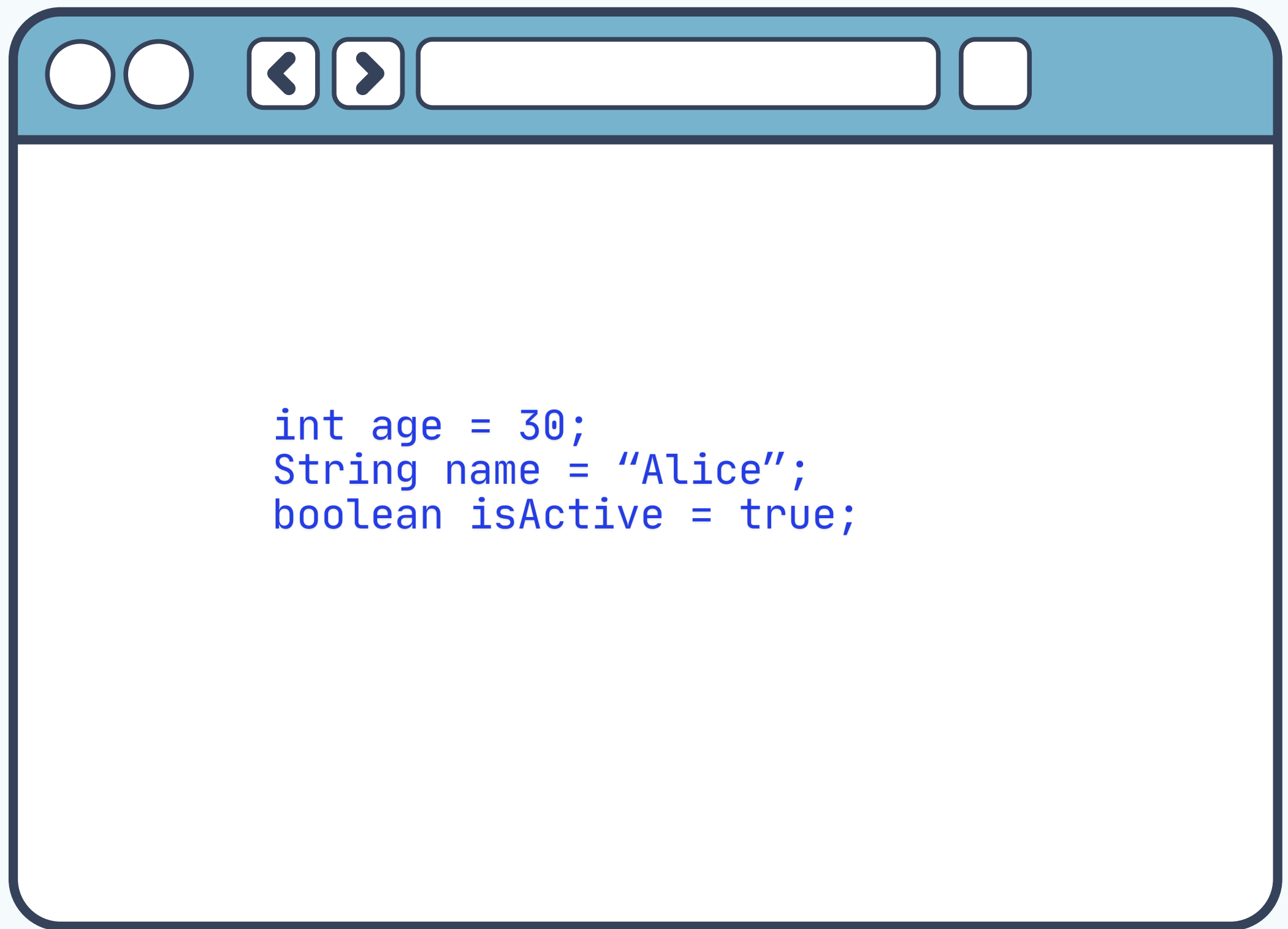
FINAL NOTE:



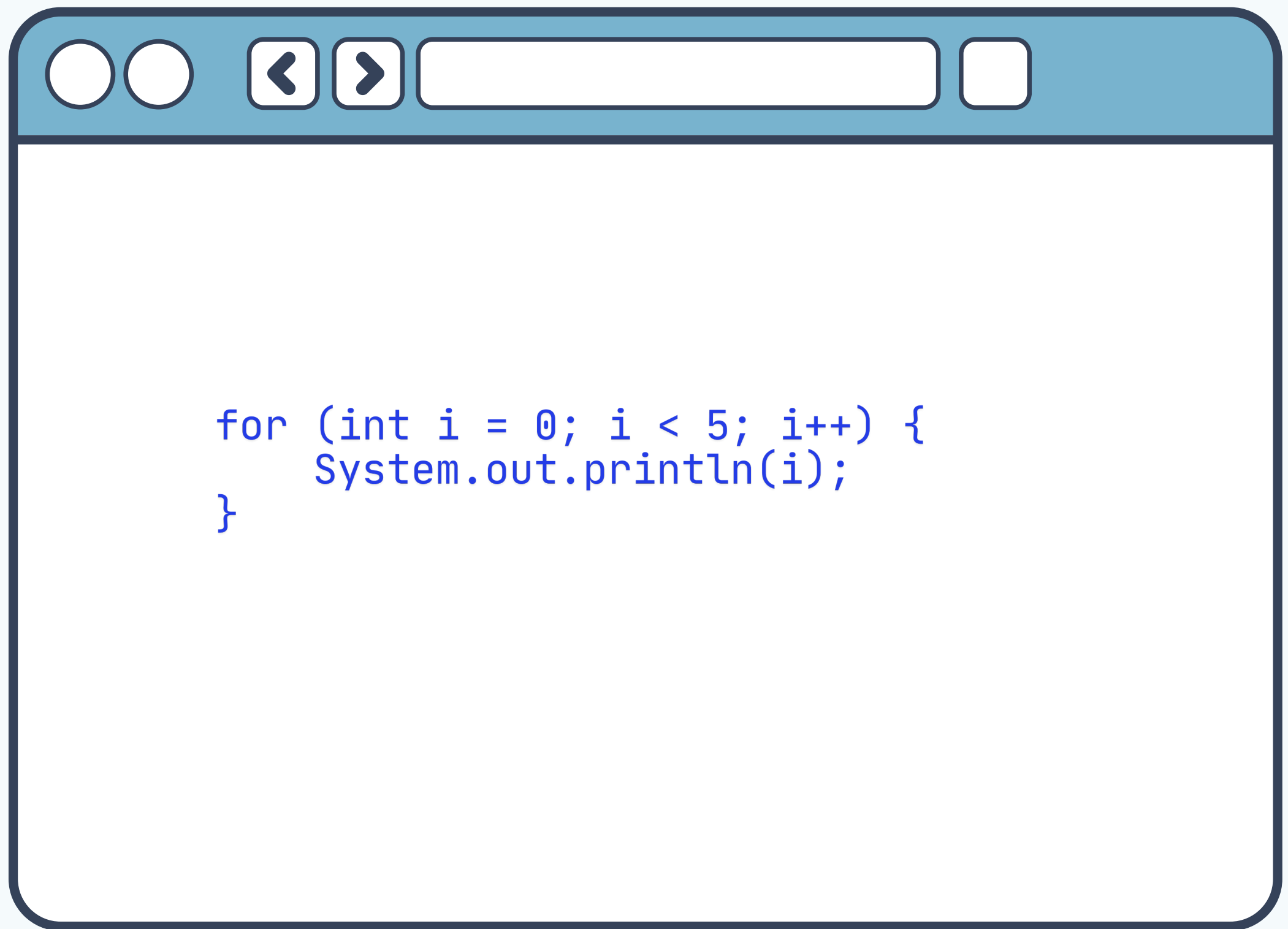
- Java is everywhere
- Master the basics and building amazing things



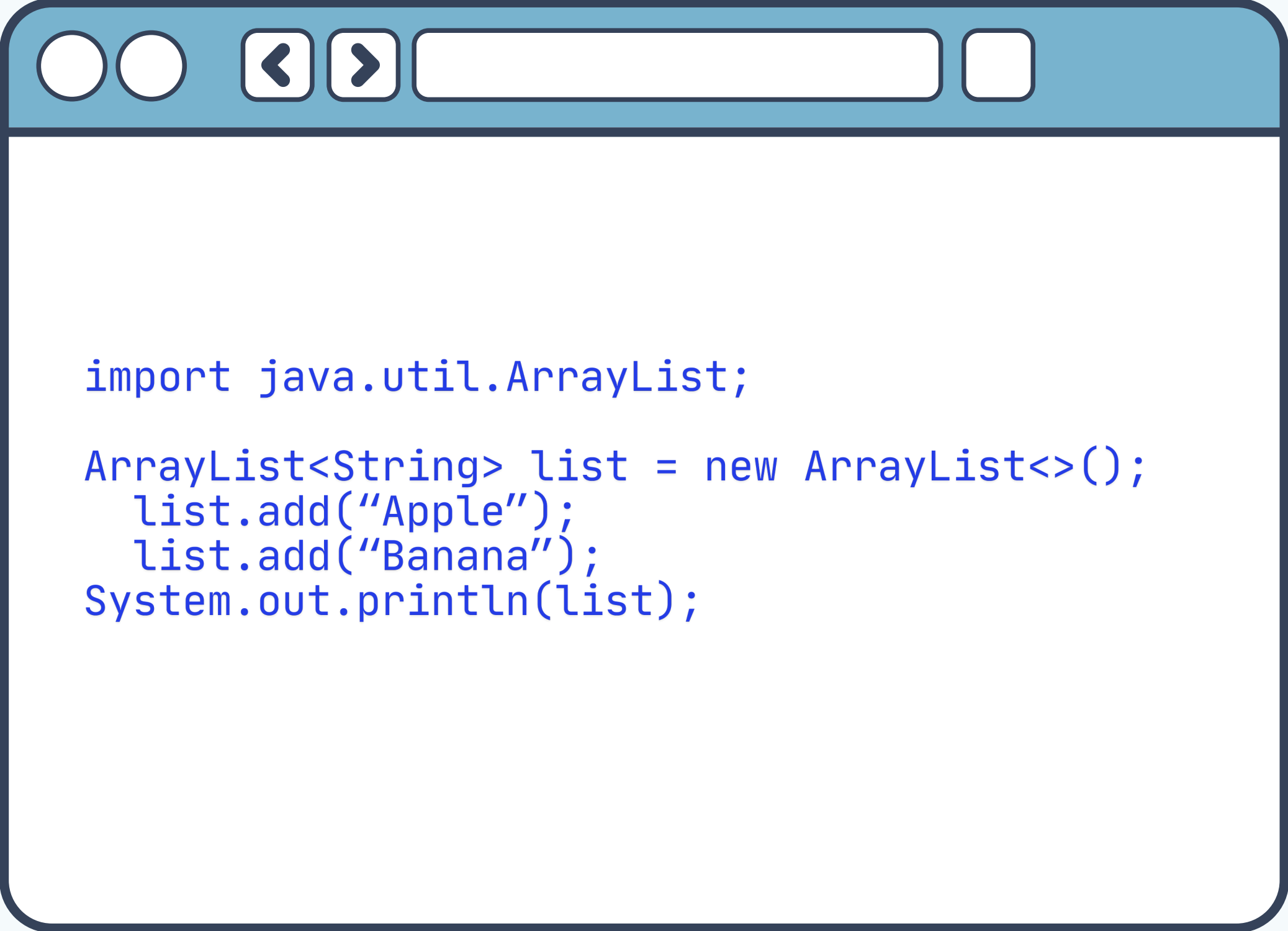
VARIABLES EXAMPLE



FOR LOOP EXAMPLE



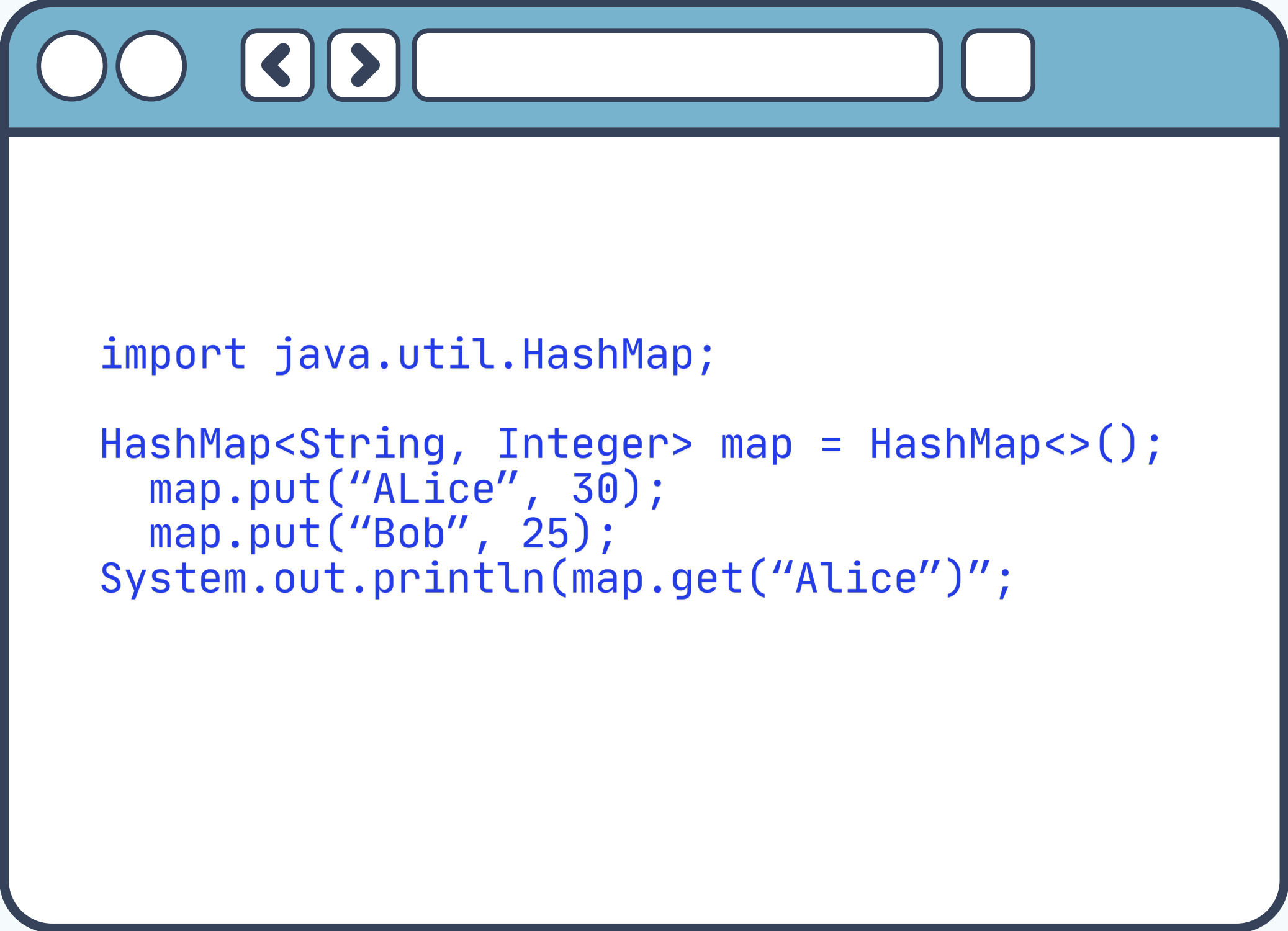
ARRAYLIST EXAMPLE



```
import java.util.ArrayList;

ArrayList<String> list = new ArrayList<>();
    list.add("Apple");
    list.add("Banana");
System.out.println(list);
```

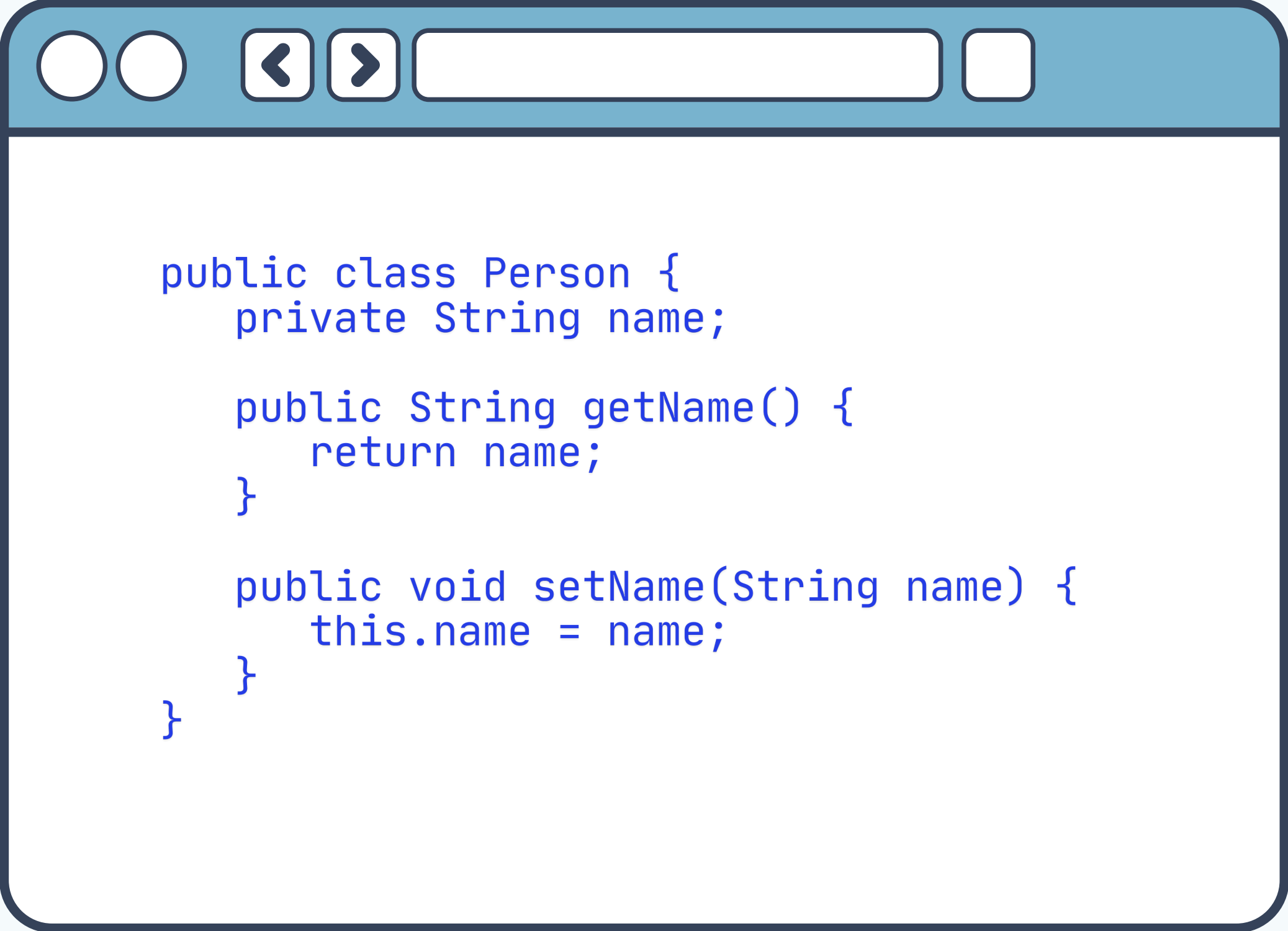
HASHMAP EXAMPLE



```
import java.util.HashMap;

HashMap<String, Integer> map = HashMap<>();
    map.put("ALice", 30);
    map.put("Bob", 25);
System.out.println(map.get("ALice"));
```

ENCAPSULATION EXAMPLE



```
public class Person {  
    private String name;  
  
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
}
```